

The Parameter Window Linear Canonical Transform: Properties and Energy Optimization

Rong-Qian Linghu and Bing-Zhao Li , *Member, IEEE*

Abstract—This paper investigates the fundamental properties of the parameter window linear canonical transform (PWLCT) and the problem of energy concentration in the time–frequency domain. Firstly, the mathematical definition of the PWLCT is established, and its fundamental properties are derived. Secondly, within the theoretical framework of the Bargmann transform, the optimization problem of time–frequency energy concentration is reformulated in Fock space, and an uncertainty principle for the PWLCT is derived. Finally, the time–frequency characteristics of Gaussian signals and nonlinear frequency modulation signals are analyzed through numerical experiments, and the Doppler characteristics of the flying heron radar echo are verified. Theoretical analysis shows that, under the normalized Fock-space formulation, disk-shaped regions provide sharp concentration bounds, and their inverse images correspond to stretched elliptical regions in the time–frequency plane. The chirp-modulated complex Gaussian window is introduced as a parameter-compatible analysis window that enables this Bargmann-Fock representation. Numerical experiments illustrate the practical concentration and tracking behavior of the proposed representation.

Index Terms—Linear canonical transform, uncertainty principle, time–frequency analysis.

I. INTRODUCTION

THE analysis of signal energy concentration in the time–frequency domain remains a central problem in modern signal processing, harmonic analysis, and quantum physics due to the uncertainty principle [1], [2]. Although Nicola and Tilli established key concentration properties for the STFT, its fixed window fundamentally limits adaptive resolution for non-stationary or structurally complex signals [3], [4], [5], [6]. The short-time linear canonical transform (STLCT) extends the STFT through adjustable kernel parameters, ensuring energy preservation, invertibility, and robustness while improving control over time–frequency structure [7], [8], [9], [10], [11]. Yet even with a Gaussian window, an essential question remains unresolved: for a region of fixed measure, which geometry maximizes energy concentration? Because the Gaussian-window STLCT inherits the fixed-window constraint [7], its adaptability

to anisotropic energy localization remains limited, highlighting the need to construct a parameter-compatible analysis window and to characterize the associated concentrating geometry under a suitable normalized phase-space representation [12], [13].

Motivated by this gap, this paper introduces the parameter window linear canonical transform (PWLCT), in which a chirp-modulated complex Gaussian window is coupled with the LCT parameters. This construction connects the PWLCT with the Bargmann-Fock framework. Through the Bargmann transform [14], [15], the energy concentration problem is reformulated in the normalized Fock space, where disk-shaped regions yield sharp concentration bounds; under the inverse PWLCT coordinate mapping, these disks correspond to stretched elliptical regions in the time–frequency plane.

The remainder of this paper is structured as follows. Section II introduces the definitions and lemmas. Section III presents the properties of the PWLCT, reformulates the energy concentration problem in the Fock space, and derives the main uncertainty principle and its potential applications. Section IV concludes the paper.

II. PRELIMINARIES

This section recalls the rearrangement lemma used in the subsequent analysis.

Lemma 1: The function μ is absolutely continuous on compact subintervals of $(0, \max u]$, and for a.e. $t \in (0, \max u)$

$$-\mu'(t) = \int_{\{u=t\}} |\nabla u|^{-1} d\mathcal{H}^1, \quad (1)$$

similarly, the function u^* is absolutely continuous on compact subintervals of $[0, +\infty)$, and for $s \geq 0$, then

$$-(u^*)'(s) = \left(\int_{\{u=u^*(s)\}} |\nabla u|^{-1} d\mathcal{H}^1 \right)^{-1}, \quad (2)$$

where $u^*(s)$ is the decreasing rearrangement function of u , u is real analytic and locally Lipschitz, $|\{|\nabla u| = 0\}| = 0$, $\partial\{u > t\} = \{u = t\}$ is smooth for a.e. $t \in (0, \max u)$, and $\mathcal{H}^1$ denotes the one-dimensional Hausdorff measure [2], [16].

III. MAIN RESULTS

A. The Parameter Window Linear Canonical Transform

Definition 1: For a function $f(t) \in L^2(\mathbb{R})$, a window function $\varphi(x) \in L^2(\mathbb{R})$, and a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad - bc = 1$, the PWLCT is defined as

$$V_f^A(x, \omega) = \int_{\mathbb{R}} f(t) \varphi(x - t) K_A(t, \omega) dt, \quad (3)$$

Received 14 May 2026; revised 4 July 2026; accepted 13 July 2026. Date of publication 17 July 2026; date of current version 29 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62171041 and in part by the Natural Science Foundation of Beijing Municipality under Grant 4242011. The associate editor coordinating the review of this article and approving it for publication was Dr. Seisuke Kyochi. (Corresponding author: Bing-Zhao Li.)

Rong-Qian Linghu is with the School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China (e-mail: lhrq0807@163.com).

Bing-Zhao Li is with the Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China (e-mail: li_bingzhao@bit.edu.cn).

Digital Object Identifier 10.1109/LSP.2026.3714739

where $K_A(t, \omega) = \frac{1}{\sqrt{|b|}} \exp\left(\frac{i\pi}{b}(at^2 - 2t\omega + d\omega^2)\right)$, $b \neq 0$, and the window $\varphi(x) = \sqrt[4]{2} \exp(-\pi\alpha x^2)$, $\alpha = 1 + \frac{a}{b}i$. Unlike the STLCT or general windowed LCT with a fixed or externally selected window, the PWLCT uses a parameter-coupled chirp-modulated complex Gaussian window. For a general chirp rate β , substitution of $\varphi_\beta(x) = \sqrt[4]{2} \exp[-\pi(1 + i\beta)x^2]$ into the LCT kernel leaves the residual quadratic phase $i\pi(a/b - \beta)t^2$. Hence, phase matching requires $\beta = a/b$, which yields the Bargmann-Fock representation. For $a = 0$ and $b = 1$, it reduces to the standard STFT Gaussian window.

B. The Properties of the PWLCT

Property 1 (Parseval's formula): For $f(t), g(t) \in L^2(\mathbb{R})$, the matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, with $ad - bc = 1$, then

$$\langle V_f^A(x, \omega), V_g^A(x, \omega) \rangle = \langle f(t), g(t) \rangle, \quad (4)$$

where $\langle f(t), g(t) \rangle = \int_{-\infty}^{\infty} f(t) \overline{g(t)} dt$.

The proof is straightforward and is omitted for brevity.

Property 2. (Inversion formula): For $f(t) \in L^2(\mathbb{R})$ can be reconstructed from the corresponding $V_f^A(x, \omega)$ via the following formula

$$f(t) = \int_{\mathbb{R}} \int_{\mathbb{R}} V_f^A(x, \omega) \overline{K_A(t, \omega) \varphi(x - t)} d\omega dx. \quad (5)$$

The proof is straightforward and is omitted for brevity.

C. The Optimization Problems to Be Solved in This Paper

Definition 2: For a function $f(t) \in L^2(\mathbb{R})$, the fraction of energy captured in a measurable subset, $\Omega \subseteq \mathbb{R}^2$, $|\Omega| = s$, then

$$\Lambda_\Omega^A = \max_{f \in L^2(\mathbb{R}) \setminus \{0\}} \frac{\int_\Omega |V_f^A(x, \omega)|^2 dx d\omega}{\int_{\mathbb{R}^2} |V_f^A(x, \omega)|^2 dx d\omega}. \quad (6)$$

This formulation yields sharp energy bounds for fixed-measure sets and their extremal geometry under the normalized Fock-space representation [3].

D. Rephrasing the Problem in the Fock Space

The Fock-space reformulation converts the anisotropic elliptical concentration problem in the time-frequency plane into a disk concentration problem in an analytic space, thereby enabling sharper concentration bounds and clearer characterization of extremal functions [3].

Lemma 2: Let the matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $z = x + i(\frac{a}{b}x + \frac{\omega}{b})$, for $f(t) \in L^2(\mathbb{R})$ the relationship between the PWLCT and the Bargmann transform [14], [15] is as follows

$$V_f^A(x, -\omega) = \frac{1}{\sqrt{|b|}} \exp\left(\pi i \frac{x\omega + d\omega^2}{b} - \frac{\pi}{2}|z|^2\right) \mathcal{B}[f(t)](z), \quad (7)$$

The Fock-space mapping is understood in normalized dimensionless time-frequency coordinates. The map z is the real linear transformation with matrix $M_A = \begin{pmatrix} 1 & 0 \\ a/b & 1/b \end{pmatrix}$, which is invertible for $b \neq 0$ and has Jacobian determinant $|\det M_A| = 1/|b|$.

Thus, the area parameter s in the Fock-space bound refers to the transformed normalized measure. The normalization is valid under $ad - bc = 1$, $b \neq 0$, $f \in L^2(\mathbb{R})$, a normalized window, and finite measurable concentration sets.

According to Definition 1 and Lemma 2, the $\Lambda_\Omega^A(f)$ can be reformulated in the following form

$$\Lambda_\Omega^A(f) = \frac{\int_\Omega |\mathcal{B}f(z)|^2 \exp(-\pi|z|^2) dm_2(z)}{\|\mathcal{B}f\|_{\mathcal{F}^2}^2}, \quad (8)$$

where $dm_2(z)$ is the Lebesgue measure on $\mathbb{C}$, and the $\mathcal{F}^2$ denote the Fock space of all entire functions. Since $\mathcal{B} : L^2(\mathbb{R}) \rightarrow \mathcal{F}^2(\mathbb{C})$ is a unitary operator, we can safely transfer the optimization problem on $\mathcal{F}^2(\mathbb{C})$, observing that

$$\Lambda_\Omega^A = \max_{F \in \mathcal{F}^2(\mathbb{C}) \setminus \{0\}} \frac{\int_\Omega |F(z)|^2 \exp(-\pi|z|^2) dm_2(z)}{\|F\|_{\mathcal{F}^2}^2}, \quad (9)$$

via the Bargmann transform, the Fock space isometrically identifies with the PWLCT representation, reducing the problem to studying squared moduli of holomorphic functions.

Theorem 1: For any measurable subset $\Omega \subset \mathbb{R}^2$, $|\Omega| = s$, we have the following inequality

$$\int_\Omega u(z) dm_2(z) \leq \int_{A_{u^*(s)}} u(z) dm_2(z), \quad (10)$$

where $A_{u^*(s)} = \{z \in \mathbb{R}^2 : u(z) > u^*(s)\}$ is the super-level set of s , $u^*(s)$ is chosen so that $|\{z : u(z) > u^*(s)\}| = s$.

Proof: Let the $t_0 = u^*(s)$, $E = \Omega \cap A_{t_0}$, $F = \Omega \setminus E$, $G = A_{t_0} \setminus E$. By dividing the set, we can split the integral into two parts $\int_\Omega u(z) dm_2(z) = \int_E u(z) dm_2(z) + \int_F u(z) dm_2(z)$, and $\int_{A_{t_0}} u(z) dm_2(z) = \int_E u(z) dm_2(z) + \int_G u(z) dm_2(z)$, according to the definition of super-level sets, $u(z) \leq t_0$, $u(z) > t_0$, then $\int_F u(z) dm_2(z) \leq t_0 |F|$, and $\int_G u(z) dm_2(z) > t_0 |G| = t_0 |F|$, therefore, we obtain $\int_F u(z) dm_2(z) \leq \int_G u(z) dm_2(z)$, according to the above integral inequality, we obtain

$$\begin{aligned} \int_\Omega u(z) dm_2(z) &= \int_E u(z) dm_2(z) + \int_F u(z) dm_2(z) \\ &\leq \int_E u(z) dm_2(z) + \int_G u(z) dm_2(z) \\ &= \int_{A_{u^*(s)}} u(z) dm_2(z). \end{aligned}$$

□

Theorem 2 (Sharp uncertainty principle for the PWLCT): For every $F \in \mathcal{F}^2(\mathbb{C}) \setminus \{0\}$ and every measurable set $\Omega \subset \mathbb{R}^2$ of finite measure, $|\Omega| = s$, we have

$$\frac{\int_\Omega |F(z)|^2 \exp(-\pi|z|^2) dm_2(z)}{\|F\|_{\mathcal{F}^2}^2} \leq 1 - \exp(-s). \quad (11)$$

The upper bound is attained, for example, by the normalized reproducing kernel in the Fock space, then $F(z) = C \exp(-\frac{\pi}{2}|z_0|^2) \exp(\pi z \bar{z}_0)$, where $C \neq 0$ and $z_0 \in \mathbb{C}$. Related extremal functions may also be characterized by a shifted Gaussian modulus. In this case, the normalized decreasing rearrangement satisfies $u^*(s) = \exp(-s)$. Correspondingly,

in the Fock space, the concentration region is the disk centered at z_0 with radius $\sqrt{s/\pi}$, whose area is exactly s .

Proof: By homogeneity, assume that $\|F\|_{\mathcal{F}^2} = 1$, and let $u(z) = |F(z)|^2 e^{-\pi|z|^2}$. Then $\int_{\mathbb{C}} u(z) dm_2(z) = 1$. By the rearrangement principle, for Ω with $m_2(\Omega) = s$,

$$\int_{\Omega} u(z) dm_2(z) \leq I(s), \quad I(s) = \int_{\{u > u^*(s)\}} u(z) dm_2(z),$$

thus, it suffices to prove $I(s) \leq 1 - e^{-s}$, for $h > 0$,

$$I(s+h) - I(s) = \int_{\{u^*(s+h) < u \leq u^*(s)\}} u(z) dm_2(z),$$

which yields $u^*(s+h)h \leq I(s+h) - I(s) \leq u^*(s)h$.

Letting $h \rightarrow 0$, we obtain $I'(s) = u^*(s)$. By Lemma 1, we obtain $-(u^*)'(s) \leq u^*(s)$, and hence $I''(s) + I'(s) \geq 0$. Therefore, for $t > s$, $I'(t) \geq I'(s)e^{s-t}$. Integrating from s to S and letting $S \rightarrow +\infty$, together with $I(S) \rightarrow 1$, gives $I'(s) + I(s) \leq 1$. Set $D(s) = e^s(1 - I(s)) - 1$. Then $D(0) = 0$ and $D'(s) = e^s(1 - I(s) - I'(s)) \geq 0$. Thus $D(s) \geq 0$, which implies $I(s) \leq 1 - e^{-s}$. Consequently,

$$\int_{\Omega} u(z) dm_2(z) \leq 1 - e^{-s}.$$

For the representative extremal case $u(z) = e^{-\pi|z-z_0|^2}$, the condition $u(z) > e^{-s}$ gives $|z - z_0| < \sqrt{\frac{s}{\pi}}$, which corresponds to a disk centered at z_0 with area s in the Fock space. $\square$

E. Potential Applications of the PWLCT

Building on theoretical analysis and experimental validation, the PWLCT provides a promising framework for non-stationary signal processing. Its adaptive window structure and Fock-space energy concentration theory support high-resolution time-frequency representation [20], [21]. Moreover, the adaptive parameter updating strategy in Algorithm 1 adjusts the PWLCT parameters according to the energy-moment characteristics of the time-frequency distribution, enabling a better match with anisotropic energy structures and complex time-varying features [22], [23], [24], [25], [26].

Algorithm 1 is a practical damped adaptive parameter-estimation scheme based on second-order energy-moment matching, rather than a globally convergent optimizer for arbitrary non-stationary signals. Under the non-degeneracy conditions $\sigma_{xx,k} > 0$, $\Delta_k = \sigma_{xx,k}\sigma_{\omega\omega,k} - \sigma_{x\omega,k}^2 > 0$, and $b > 0$, the shape-matching condition uniquely determines the unregularized pair (a_k^*, b_k^*) . Once $d = d_0$ is fixed, c is determined by $ad - bc = 1$. The convergence rate, local/global convergence conditions, and robustness under general signal models remain important topics for future work.

1) *Energy Capture at Fixed Area:* The input signal is

$$f(t) = c \cdot \exp(i\omega_0 t) \cdot \exp\left(-\frac{(t-x_0)^2}{2\sigma^2}\right), \quad (12)$$

in this paper, we use a Gaussian test signal $f(t)$ sampled at 197 Hz over 6 seconds (time from 0 to 5.9797s), with unit amplitude, center frequency 50 Hz (angular frequency 100π rad/s), temporal center at 3 s, and a Gaussian envelope with standard deviation 0.19.

Algorithm 1: Adaptive Parameter Updating Method.

Input: Signal $f(t)$, grid (x, ω) , initial matrix A_0 , fixed parameter d_0 , step size η , tolerance δ , maximum iteration $K_{\max}$

Output: Updated PWLCT parameter matrix A^*

1: **for** $k = 0, 1, \dots, K_{\max} - 1$ **do**

2: Compute the PWLCT energy distribution

$$P_k(x, \omega) = |V_{A_k} f(x, \omega)|^2.$$

3: Compute the weighted centroid and centralized coordinates

$$\bar{x}_k = \frac{\sum x P_k}{\sum P_k}, \quad \bar{\omega}_k = \frac{\sum \omega P_k}{\sum P_k}, \quad X = x - \bar{x}_k, \quad Y = \omega - \bar{\omega}_k.$$

4: Compute the second-order energy moments

$$\sigma_{xx,k} = \frac{\sum P_k X^2}{\sum P_k}, \quad \sigma_{x\omega,k} = \frac{\sum P_k XY}{\sum P_k}, \quad \sigma_{\omega\omega,k} = \frac{\sum P_k Y^2}{\sum P_k}.$$

5: Estimate the energy-concentration guided parameters. Let

$$\Sigma_k = \begin{pmatrix} \sigma_{xx,k} & \sigma_{x\omega,k} \\ \sigma_{x\omega,k} & \sigma_{\omega\omega,k} \end{pmatrix}.$$

6: Under the local elliptically contoured energy model

$p_k(r) = \rho_k(r^T \Sigma_k^{-1} r)$, where $r = (X, Y)^T$ and ρ_k is strictly decreasing, the fixed-area optimal energy concentration region is the super-level ellipse

$r^T \Sigma_k^{-1} r \leq \gamma$. Meanwhile, the inverse image of a

Fock-space disk under the PWLCT coordinate

$z = x + i(\frac{a}{b}x + \frac{\omega}{b})$ is $r^T T(a, b)^T T(a, b) r \leq R^2$, where

$$T(a, b) = \begin{pmatrix} 1 & 0 \\ a/b & 1/b \end{pmatrix}. \text{ Thus, matching the}$$

PWLCT-induced ellipse with the optimal super-level

energy ellipse gives $T(a, b)^T T(a, b) = \lambda \Sigma_k^{-1}$. Solving this isotropy condition gives

$$\hat{b}_k = \sqrt{\max \left[\frac{(\sigma_{xx,k} + \epsilon)(\sigma_{\omega\omega,k} + \epsilon) - \sigma_{x\omega,k}^2}{(\sigma_{xx,k} + \epsilon)^2}, \epsilon_b \right]}.$$

where ϵ and ϵ_b are small positive constants used to avoid division by zero and ensure $b_k > 0$.

7: Update the parameters and reconstruct the LCT matrix

$$a_{k+1} = (1 - \eta)a_k + \eta\hat{a}_k, \quad b_{k+1} = (1 - \eta)b_k + \eta\hat{b}_k,$$

$$d_{k+1} = d_0, \quad c_{k+1} = \frac{a_{k+1}d_0 - 1}{b_{k+1}}, \quad A_{k+1} = \begin{pmatrix} a_{k+1} & b_{k+1} \\ c_{k+1} & d_0 \end{pmatrix}.$$

8: **if** $\|A_{k+1} - A_k\|_F < \delta$ **then**

9: **break**

10: **end if**

11: **end for**

12: **return** $A^* = A_{k+1}$.

In Fig. 1, after the PWLCT is applied, the Gaussian signal $f(t)$ exhibits an elliptical energy distribution on the time-frequency plane centered at (3.0 s, 50 Hz) with a peak of 0.018.

Fig. 2 shows that, for equal-area regions, the elliptical disk achieves higher energy-capture efficiency than the circular disk, with its energy proportion curve closer to the theoretical bound $1 - e^{-s}$.

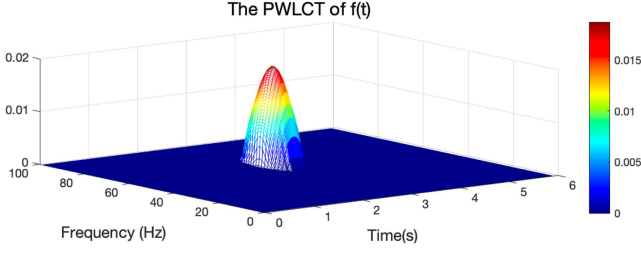
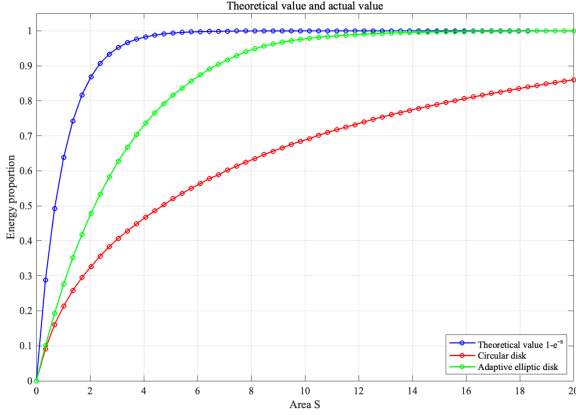
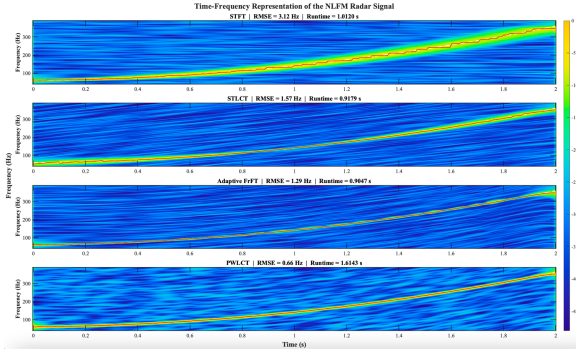
Fig. 1. Time-frequency representation of signal $f(t)$.Fig. 2. Comparison of energy capture ratio between an elliptical disk (PWLCT, $A = [-0.77, 0.28; -3.57, 0]$) and a circular disk (STFT, $A = [0, 1; -1, 0]$) under area S .

Fig. 3. Comparative time-frequency representations obtained by STFT, STLCT, adaptive FrFT, and PWLCT.

2) *Time-Frequency Analysis of Nonlinear Frequency-Modulated (NLFM) Signal*: The input signal is

$$g(t) = \exp \left\{ j2\pi \left[f_0 t + H \left(0.04 \frac{t^2}{T} + \frac{0.92}{3} \frac{t^3}{T^2} \right) \right] \right\}, \quad (13)$$

where $g(t)$ denotes the complex NLFM signal, t is the time variable, T is the signal duration, f_0 is the initial frequency, f_1 is the final frequency, and $H = f_1 - f_0$ denotes the signal bandwidth. The term t/T represents the normalized time variable, which is dimensionless and is used to define the nonlinear frequency-modulation law. In this experiment, the parameters are set as $f_0 = 60$ Hz, $f_1 = 360$ Hz, $H = 300$ Hz, and $T = 2$ s.

Fig. 3 compares the time-frequency representations of the NLFM signal obtained by STFT, STLCT, adaptive fractional

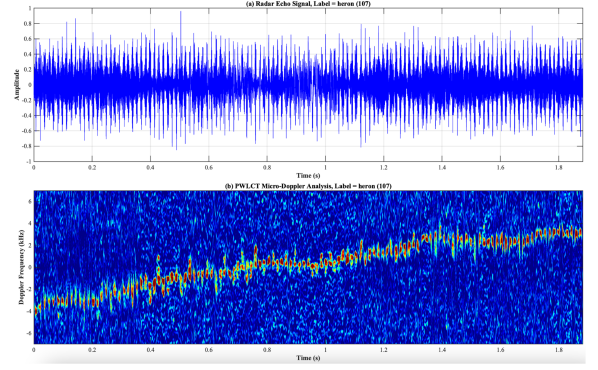


Fig. 4. Radar echo Doppler analysis of a flying heron.

Fourier transform (FrFT) [29], and PWLCT. The STFT shows obvious energy spreading and has the largest root-mean-square error (RMSE) of 3.12 Hz, while the STLCT and FrFT reduce the RMSE to 1.57 Hz and 1.29 Hz, respectively. In comparison, the PWLCT achieves the smallest RMSE of 0.66 Hz, indicating that its adaptive parameter updating mechanism better matches the local time-frequency structure of the NLFM signal. However, its runtime increases to 1.6143 s because of the additional iterative parameter updating process, reflecting a trade-off between tracking accuracy and computational cost.

3) *Doppler Analysis of Flying Heron Radar Echoes*: To validate the applicability of the PWLCT method to real radar signals, this study uses the radar echo data from the third range cell of the 107th heron measurement in the public dataset [27], [28] “Radar measurements on drones, birds and humans with a 77 GHz FMCW sensor” as the analysis object.

Fig. 4 presents the radar echo from the third range cell of the heron target along with its PWLCT-based Doppler analysis results. In Fig. 4(a), the preprocessed time-domain echo exhibits dense oscillations and random fluctuations, making it difficult to directly identify target motion characteristics. In contrast, the PWLCT time-frequency representation shown in Fig. 4(b) clearly extracts the dominant Doppler ridge, which smoothly transitions from negative to positive frequencies, reflecting the time-varying radial motion state of the target. Furthermore, the weak fluctuations and local energy spreading around the main ridge may be related to micro-Doppler modulation induced by wing flapping.

IV. CONCLUSION

This paper introduces the PWLCT as a parameter-compatible representation for studying time-frequency energy concentration. By coupling a chirp-modulated complex Gaussian window with the LCT parameters, the proposed transform provides a flexible representation for anisotropic time-frequency energy distributions. Through the Bargmann transform, the energy concentration problem is reformulated in the normalized Fock space, where disk-shaped regions provide sharp concentration bounds. Under the inverse PWLCT coordinate mapping, these disks correspond to stretched elliptical regions in the time-frequency plane. Numerical examples illustrate the practical energy concentration and tracking behavior of the proposed representation for Gaussian, NLFM, and radar echo signals.

REFERENCES

- [1] J. Zhong and Y. Huang, "Time-frequency representation based on an adaptive short-time Fourier transform," *IEEE Trans. Signal Process.*, vol. 58, no. 10, pp. 5118–5128, Oct. 2010.
- [2] M. Garrido, "The feedforward short-time Fourier transform," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 63, no. 9, pp. 868–872, Sep. 2016.
- [3] F. Nicola and P. Tilli, "The Faber–Krahn inequality for the short-time Fourier transform," *Invent. Math.*, vol. 230, no. 1, pp. 1–30, 2022.
- [4] F. Marceca and J. L. Romero, "Spectral deviation of concentration operators for the short-time Fourier transform," *Stud. Math.*, vol. 270, no. 2, pp. 145–173, 2023.
- [5] Z. C. Zhang and X. Shi, "Kernel function- τ -Wigner distribution associated with the linear canonical transform," *IEEE Signal Process. Lett.*, vol. 29, pp. 1764–1768, 2022.
- [6] Y. Mo, T. Qian, W. X. Mai, and Q. H. Chen, "The AFD methods to compute Hilbert transform," *Appl. Math. Lett.*, vol. 45, pp. 18–24, 2015.
- [7] K. I. Kou and R. H. Xu, "Windowed linear canonical transform and its applications," *Signal Process.*, vol. 92, no. 1, pp. 179–188, 2012.
- [8] Z. C. Zhang, "Sampling theorem for the short-time linear canonical transform and its applications," *Signal Process.*, vol. 113, pp. 138–146, 2015.
- [9] Y. N. Sun and B. Z. Li, "Sliding discrete linear canonical transform," *IEEE Trans. Signal Process.*, vol. 66, no. 17, pp. 4553–4563, Sep. 2018.
- [10] D. Y. Wei and H. Hu, "Theory and applications of short-time linear canonical transform," *Digit. Signal Process.*, vol. 118, 2022, Art. no. 103239.
- [11] G. Huang, F. Zhang, and R. Tao, "Sliding short-time fractional Fourier transform," *IEEE Signal Process. Lett.*, vol. 29, pp. 1823–1827, 2022.
- [12] L. D. Abreu, J. A. M. Pereira, and J. L. Romero, "Sharp rates of convergence for accumulated spectrograms," *Inverse Problems*, vol. 33, no. 11, 2017, Art. no. 115008.
- [13] D. Bucur and G. Buttazzo, *Variational Methods in Shape Optimization Problems*. Boston, MA, USA: Birkhauser, 2005.
- [14] K. Zhu, *Analysis on Fock Spaces*. New York, NY, USA: Springer, 2012.
- [15] V. Bargmann, "On a Hilbert space of analytic functions and an associated integral transform, part II," *Commun. Pure Appl. Math.*, vol. 20, pp. 1–101, 1967.
- [16] A. Baernstein II, D. Drasin, and R. S. Laugesen, *Symmetrization in Analysis, New Mathematical Monographs*, vol. 36. Cambridge, U.K.: Cambridge Univ. Press, 2019.
- [17] F. A. Berezin, "Wick and anti-wick symbols of operators," *Mat. Sb (N.S.)*, vol. 86, no. 128, pp. 578–610, 1971.
- [18] I. Daubechies, "Time-frequency localization operators: A geometric phase space approach," *IEEE Trans. Inf. Theory*, vol. 34, no. 4, pp. 605–612, Jul. 1988.
- [19] M. W. Wong, *Wavelet Transforms and Localization Operators, Operator Theory: Advances and Applications*, vol. 136. Basel, Switzerland: Birkhauser Verlag, 2002.
- [20] J. Shi, G. Chen, Y. Zhao, and R. Tao, "Synchrosqueezed fractional wavelet transform: A new high-resolution time-frequency representation," *IEEE Trans. Signal Process.*, vol. 71, pp. 264–278, 2023.
- [21] J. Shi, X. Liu, X. Fang, X. Sha, W. Xiang, and Q. Zhang, "Linear canonical matched filter: Theory, design, and applications," *IEEE Trans. Signal Process.*, vol. 66, no. 24, pp. 6404–6417, Dec. 2018.
- [22] G. Yu and Y. Zhou, "General linear chirplet transform," *Mech. Syst. Signal Process.*, vol. 70–71, pp. 958–973, 2016.
- [23] S. Chen, Y. Yang, Z. Peng, X. Dong, W. Zhang, and G. Meng, "Adaptive chirp mode pursuit: Algorithm and applications," *Mech. Syst. Signal Process.*, vol. 116, pp. 566–584, 2019.
- [24] J. Zhang, Y. Bao, T. Aoki, and T. Yamashita, "Refined linear chirplet transform for time–frequency analysis of non-stationary signals," *Mech. Syst. Signal Process.*, vol. 223, 2025, Art. no. 111785.
- [25] S. G. Mallat and Z. Zhang, "Matching pursuits with time-frequency dictionaries," *IEEE Trans. Signal Process.*, vol. 41, no. 12, pp. 3397–3415, Dec. 1993.
- [26] S. Mann and S. Haykin, "Adaptive chirplet transform: An adaptive generalization of the wavelet transform," *Opt. Eng.*, vol. 31, no. 6, pp. 1243–1256, 1992.
- [27] A. Karlsson, "Radar measurements on drones, birds and humans with a 77 GHz FMCW sensor," Zenodo Dataset, Version v2.0, Sep. 16, 2021. [Online]. Available: <https://zenodo.org/records/5896641>
- [28] A. Karlsson, M. Jansson, and M. Hamalainen, "Model-aided drone classification using convolutional neural networks," in *Proc. IEEE Radar Conf.*, 2022, pp. 1–6.
- [29] J. Guan, "Adaptive fractional Fourier transform-based detection algorithm for moving target in heavy sea clutter," *IET Radar, Sonar Navigation*, vol. 6, no. 5, pp. 389–401, 2012.